ECE501-2025-ProjectNo16-GroupNo4

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Abstract

Massive amounts of data are produced by medical imaging, necessitating effective transmission and storage without sacrificing diagnostic value. The theoretical underpinnings of picture partitioning for lossless JPEG compression-based compact encoding are presented in this research. Redundancy is decreased and important visual details are maintained by splitting an image into several rectangular segments and encoding each one separately. High compression ratios appropriate for medical applications where precision is crucial are the goal of the method.

1 Introduction

There is an exponential growth in the technologies of medical imaging. Technologies like x-rays, MRI and CT scans have led to a rise in the amount of images generated daily. Since these medical images frequently include important diagnostic data, they must be sent and stored without sacrificing image quality or data. However, effectively managing massive amounts of data while preserving diagnostic accuracy presents significant hurdles for medical organizations.Image compression methods are used to solve this problem by lowering the file size while maintaining necessary data. Among several tactics, picture partitioning for compact encoding has proven to be a successful method for preserving the integrity of medical images while maximizing compression speed.

2 Approach

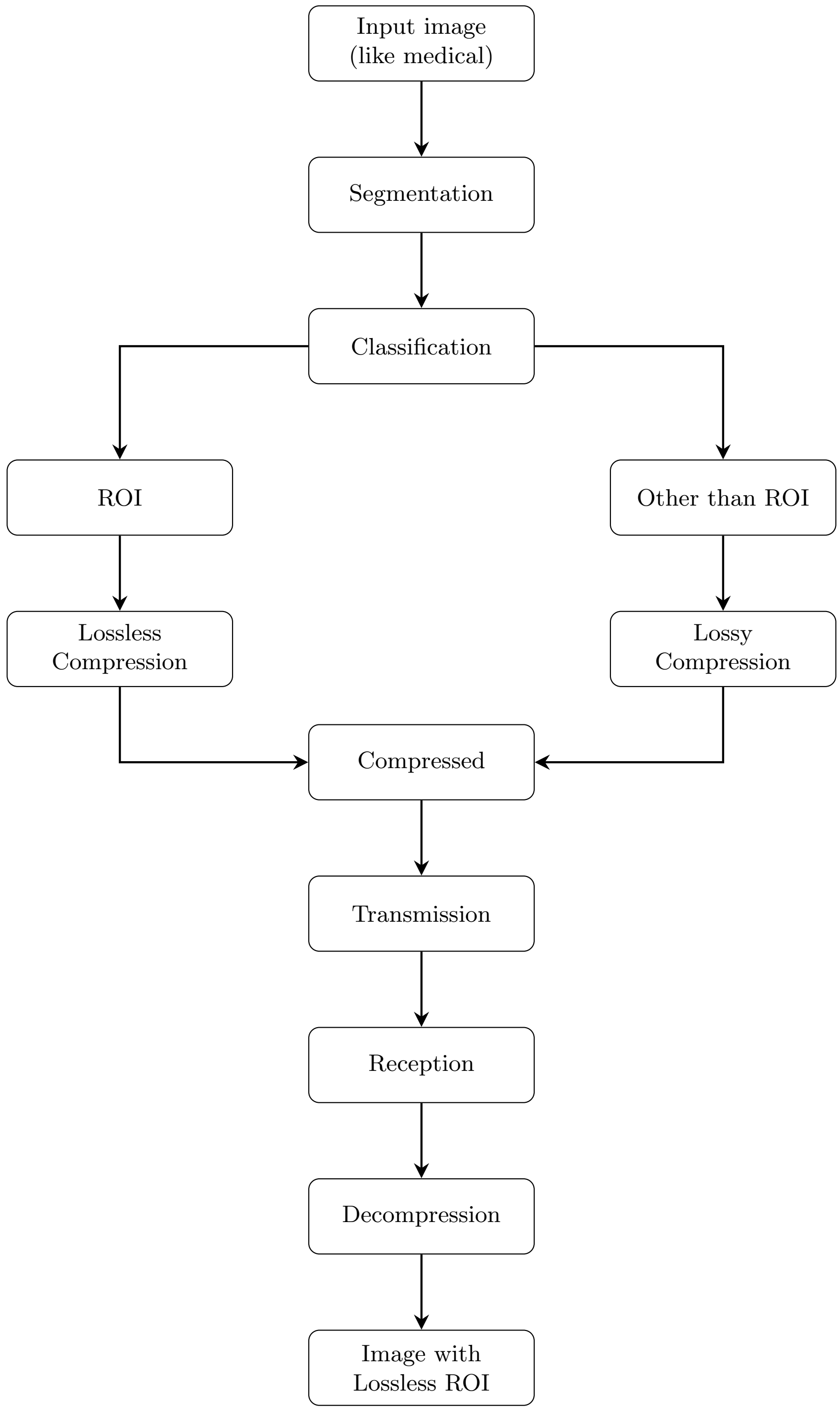

Figure 1: Block diagram of the proposed method.

2.1 Explanation of the Proposed Method

- **Medical Image (CT/MRI):**
  The process begins with the input of a medical image, such as a CT or MRI scan.

- **Segmentation:**
  The image is segmented to isolate different regions.

- **Classification:**
  Segmented regions are classified into two categories:

  - **ROI (Region of Interest):** Critical regions for diagnosis.
  - **Other than ROI:** Background or less important areas.

- **Lossless Compression of ROI:**
  The ROI is compressed using lossless methods to preserve all diagnostic information.

- **Lossy Compression of Other Regions:**
  Non-critical areas are compressed using lossy techniques to reduce data size.

- **Compressed:**
  Both lossless and lossy compressed data are combined into a unified compressed image.

- **Transmission:**
  The compressed image is transmitted to the desired location or system.

- **Reception:**
  The system receives the transmitted compressed image data.

- **Decompression:**
  The image is decompressed. ROI is reconstructed without any loss, and non-ROI with acceptable loss.

- **Image with Lossless ROI:**
  Final output image is restored with complete quality in ROI and lossy in other regions.

3 Importance of it in Medical Imaging

The great resolution and diagnostic value of medical photographs define them. The interpretation of clinical outcomes might be impacted by even the slightest data loss or distortion. Therefore, in medical applications, lossless compression is favored over lossy techniques. By maintaining all pixel information, lossless compression guarantees that the decompressed image is an identical duplicate of the original.

However, medical content is diverse, straight compression of huge medical images can be computationally costly and result in less-than-ideal compression ratios. By dividing the picture into smaller sections, the compression algorithm may adjust to the unique features of each area:

- Higher compression can be used to encode homogeneous areas (such as background tissues).
- To maintain diagnostic information, high-detail areas (such as tumors or the borders of organs) can be properly encoded.

Partitioning thus aids in striking a compromise between information preservation and compression efficiency, two essential needs in medical image processing.

4 Next Week Plan

In the upcoming week, we will focus on the following tasks:

- Identify and study the algorithms corresponding to the given approaches.
- Explore and analyze the selected research papers in greater detail.
- Review additional references to strengthen the understanding of the proposed methods.